LEXIGRID

SMTP and Distributed Notification Workers

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Shahryar Jahangir, Aniza Inc.

Procedures, examples, verification and operational limits. Complete commissioning on your deployment before serving customer traffic. Sample values are fictional; original keys and data must be preserved.

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LexiGrid 5.54.15 - SMTP Configuration and Distributed Workers

Author: Shahryar Jahangir, Aniza Inc.

Purpose and architecture

LexiGrid provides browser-based SMTP administration and concurrent risk notification processing. SMTP settings are global, not tenant-specific. Only platform administrators may change them. Tenant administrators may inspect status and test connectivity.

Prerequisites

A stable 32-byte base64url LEXIGRID_MAIL_CONFIG_KEY must be configured identically on every application instance. Compose passes the existing key and the SMTP environment defaults to both the API and workflow scheduler. Configure LEXIGRID_SMTP_USERNAME and LEXIGRID_SMTP_SECURITY in the Compose environment. The older LEXIGRID_SMTP_USER value remains a fallback for the username. The installer generates the key for new native installations and adds it to existing native configuration files if missing. Back up the key securely. Losing it makes previously saved SMTP overrides unreadable.

Step 1 - Open the administration page

Sign in as platform administrator. Open Email Configuration. The page displays the active SMTP host, port, security mode, sender, configuration revision, and whether credentials are present. Password values are never returned to the browser.

Step 2 - Enter SMTP settings

Enter the SMTP host (for example mail.example.com), port (usually 587 for STARTTLS or 465 for implicit TLS), security mode, sender address, and SMTP username. Enter a new password only when changing it; leave the password blank to retain the current one. Select Clear existing password to remove it. Save configuration.

Step 3 - Test connectivity

Select Test SMTP connection and authentication. The test opens a TLS-protected SMTP connection, authenticates if configured, and performs NOOP. It does not send an email. Send a risk update to a consenting test recipient to validate actual delivery.

Example: configured SMTP

Host: mail.example.com | Port: 587 | Security: STARTTLS | From: lexigrid@example.com | Username: mailer@example.com | Password: entered through the protected form and encrypted at rest. The page does not display the password.

Settings precedence

Browser-managed SMTP settings are encrypted in the shared database and override the corresponding environment settings. Environment settings provide fallback defaults. Changes take effect on the next email send; restarting is not required for SMTP changes. LEXIGRID_MAIL_WORKER is deployment configuration and changing it requires restarting the worker process.

Distributed worker deployment

For multiple worker processes, use a shared PostgreSQL database and set LEXIGRID_MAIL_WORKER=1 on each designated worker instance. The workers claim pending notifications using SELECT FOR UPDATE SKIP LOCKED. A claim is leased for 10 minutes. Other workers can continue processing different messages. Each worker must use the same SMTP encryption key and database.

SQLite behavior

SQLite supports atomic compare-and-swap claiming for local concurrent processes but is not recommended for multi-host deployment. Use PostgreSQL for a production distributed-worker topology. SMTP delivery is at-least-once: a crash after SMTP acceptance but before status update may result in duplicate email.

Scheduled reminders

Reminder records are claimed atomically. Each reminder is advanced to its next due time within the same transaction that queues its risk updates. Closed risks are skipped. Reminder timing depends on a running worker; the worker polls approximately every 60 seconds.

Security boundaries

SMTP credentials are encrypted at rest using AES-256-GCM and a dedicated deployment key. The API never returns the password. Changes require the platform administrator role and are audited. All application instances must use trusted TLS and restricted database credentials. Protect backups and database snapshots; a copy of the encryption key plus database permits decryption.

Operational limitations

The browser cannot activate a stopped worker process: set LEXIGRID_MAIL_WORKER=1 in deployment configuration and restart. This release does not implement distributed leadership election, an external queue, or guaranteed exactly-once email delivery. Provider delivery receipts still require a separately configured signed webhook adapter.

Troubleshooting

SMTP ready=false: inspect validation errors and check host, sender, and credentials. Test connection fails: check DNS, firewall, SMTP TLS mode, provider authentication, and restricted application logs. Worker disabled: enable the worker in deployment config and restart. Key error: confirm the same original LEXIGRID_MAIL_CONFIG_KEY is present on all nodes; never replace an active key without a controlled rotation process.

Upgrade procedure

Back up the database and current .env/.env.native. Upgrade using the installer and preserve database/configuration. After a verified backup/restore drill, apply migrations to the schema required by the installed distribution using the schema-owner connection. Use restricted runtime credentials for the API/workers. Verify the new mail encryption key is present. Restart the application, open Email Configuration, save SMTP settings, test connectivity, and verify a notification from a test risk.

Verification and acceptance criteria

Confirm platform administrator can save SMTP changes; tenant administrator cannot. Confirm the password is absent from API responses and database plaintext. Confirm stale revision is rejected. Confirm two workers process distinct pending messages under PostgreSQL row locking and that reminders do not double-enqueue. Confirm failed SMTP sends retry and appear in notification history.

Workflow channel recovery

Workflow delivery records are tenant-linked and channel-specific. Five automatic attempts use 2/4/8/16-minute backoff and ten-minute reclaimable leases. Stale workers cannot finish a newer claim. Failed channels are listed in Workflows; tenant administrators/risk managers may request one additional attempt with a reason and current attempt count. Platform administrators act only within their selected tenant. A concurrent/stale request returns 409. Attempts remain cumulative, and successful channels are not retried because another channel failed. Stable webhook Idempotency-Key and email Message-ID support receiver deduplication; crash-boundary duplication remains possible.

Workflow decisions and delivery recovery

Review a governed record

Create/publish a workflow definition with designated approvers and optional due dates. Start a run for the exact record and target state. A run snapshots approver steps and the reviewed content digest/version. The initiator cannot approve their own run, and only the assigned approver can decide a step. A stale version or resolved run is a conflict, not permission to repeat an approval.

After required decisions complete, retry the governed transition. Approval consumption and transition are atomic and single-use. Changing material record content invalidates the prior approval. An unrelated/legacy approval does not authorize the transition. Check decision history for the actual step and actor, not merely the final state label.

Inbox and overdue checks

The workflow inbox shows the user's own notifications and channel status. Mark read records reading in the application; it is not SMTP delivery confirmation. Tenant administrators may trigger overdue checks. The independent scheduler runs periodic escalation/delivery; native startup alone does not install that scheduler.

Channel retry and manual recovery

Enabled webhook/email channels have separate durable states: queued, sending, retry, retry_requested, failed or delivered. Automatic delivery stops after five attempts, using 2/4/8/16-minute failure backoff and a ten-minute lease. A successful channel is not resent when another channel fails. Disabled channels keep their state until re-enabled.

After correcting the transport issue, a permitted tenant administrator/risk manager may use Failed workflow deliveries to request one additional attempt with a reason and current attempts. Platform administrators act within the selected tenant. A stale/concurrent request returns 409; refresh first. Attempts never reset, and the retry event is audited in the same transaction.

An expired sending lease can mean remote acceptance already occurred. Message-ID/Idempotency-Key stays stable, but the guarantee is at least once. Use receiver deduplication and inspect gateway evidence before manually retrying a suspected accepted message. Do not delete failures or rewrite audit history to hide retries.

Exact content and author independence

, creators, recorded editors and identified owners cannot approve the target through another workflow initiator. Each approval step rechecks the reviewed content digest/version before deciding, in addition to the existing check before consumption. Reject a stale pending run and start a fresh review. Native risk/policy/finding requests also carry exact-content bindings; stale or legacy unbound approvals cannot authorize new actions. Risk acceptance has its own independent decision and mandatory UTC expiry in the Risk Acceptance workspace.